
EECS 16A Imaging 1

TA, UCSI, UCSI, UCSI

DSP Accommodations

- If you need any DSP accommodations (lab-specific or general class-related), please make sure your accommodation letter has been submitted.
- If you need anything further regarding lab, please reach out to Anastasia and Carol at the lab email:
eeecs16a.lab@berkeley.edu

Semester Outline



Imaging
Module



Touchscreen
Module



Acoustic
Positioning
Module

Why Imaging?

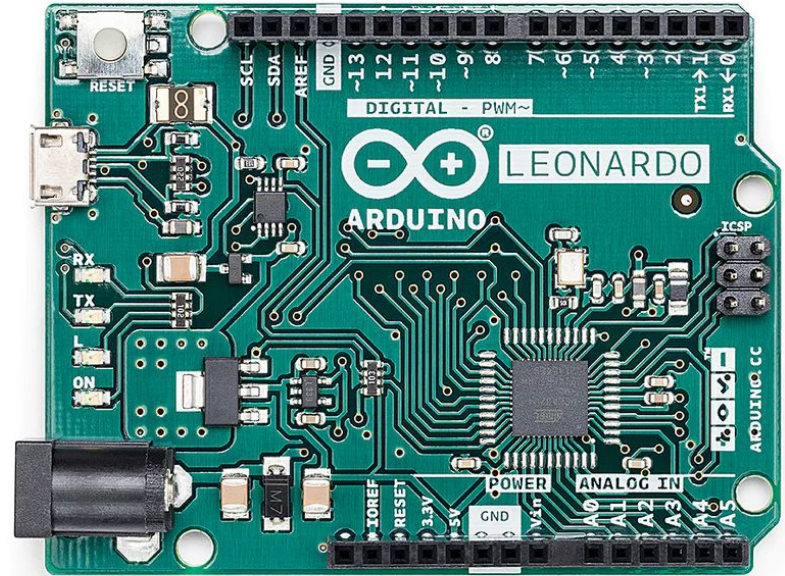
- **Module Idea:** use linear algebra techniques to capture real world images with limited sensors.
- **Today:**
 - No linear algebra yet!
 - First, we need to find a link between physical quantities (light) and voltage.
 - If you can digitize physical world, you can do anything (IOT devices, internet, code, processing).
 - Digitizing - making information about physical world processable by a computer.

Today's Lab: Imaging Part 1

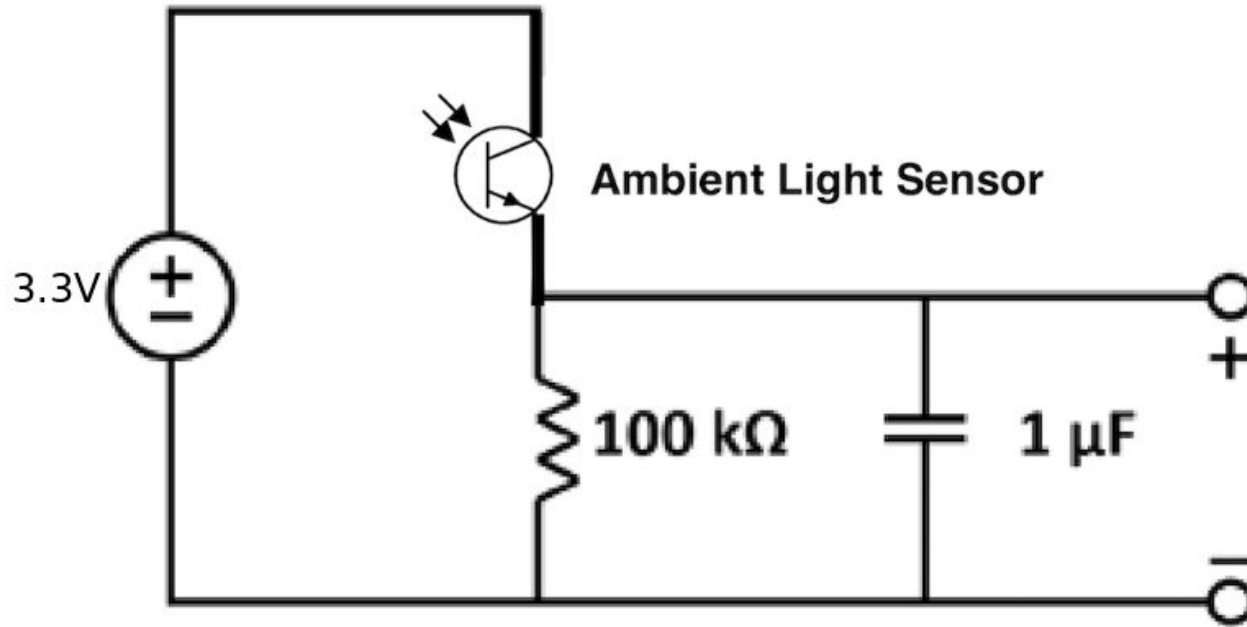
- You'll receive lab kit materials after completing part 1 of today's lab
- Circuits + Breadboarding 101
- Build circuit that reacts to light intensity
 - Use Arduino Leonardo microcontroller (+ Oscilloscope) to see how the circuit behaves
- Please read the Jupyter Notebook carefully – the course staff won't answer questions that are clearly answered in the notebook.
- **Graded checkoff starts today!**

Arduino Leonardo

- A microcontroller board
- Logistics:
 - Distributed by lab assistants at the beginning of each lab
 - 1 board per student pair
 - You **must return the board and the cable** during the check-off or at the end of the lab section, whichever comes first. Otherwise, you won't get credit for the lab.

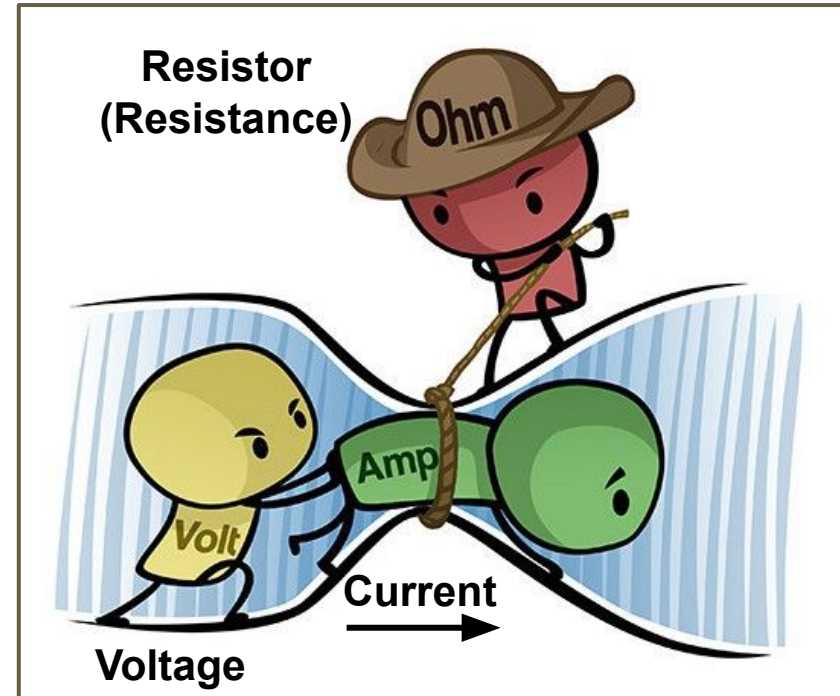


Our circuit



A Little Physics: Voltage, Current, and Resistors

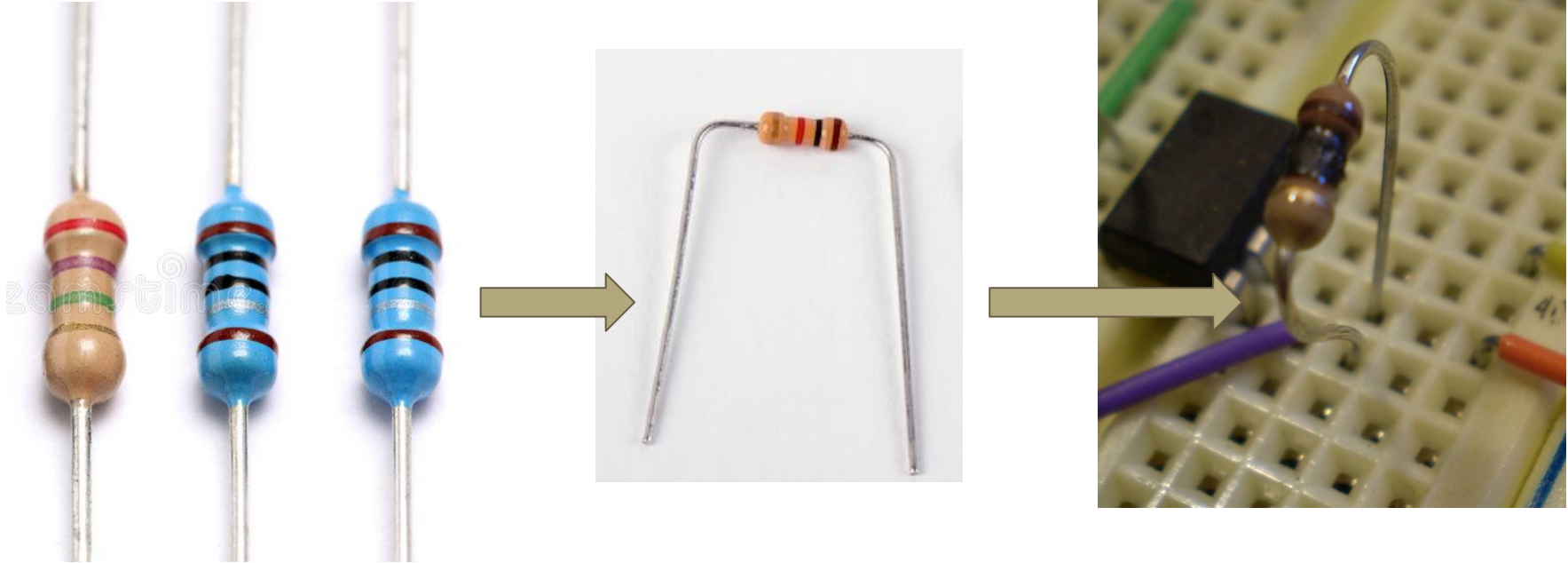
- **Voltage [Volts]** - electric “pressure” that pushes charge through circuit
- **Current [Amps]** - flow of charge through circuit
 - 1 Amp = 1 Coulomb per second =
- **Resistor [Ohms]** - circuit component that resists the flow of charge through circuit



Simple Circuit: The Tools™

- Components
 - Resistors
 - Capacitors
 - Voltage Source
- Wires / Jumpers
 - pins vs. sockets

What's in your circuit? : Resistors



What's on your circuit? : Resistors



4 Band Resistor Color Coding				
COLOR	1ST BAND	2ND BAND	MULTIPLIER	TOLERANCE
BLACK	0	0	$\times 1\Omega$	
BROWN	1	1	$\times 10\Omega$	$\pm 1\%$
RED	2	2	$\times 100\Omega$	$\pm 2\%$
ORANGE	3	3	$\times 1000\Omega$	
YELLOW	4	4	$\times 10000\Omega$	
GREEN	5	5	$\times 100000\Omega$	$\pm 0.5\%$
BLUE	6	6	$\times 1000000\Omega$	± 0.25
VIOLET	7	7	$\times 10000000\Omega$	± 0.10
GREY	8	8		± 0.05
WHITE	9	9		
GOLD			0.1	$\pm 5\%$
SILVER			0.01	$\pm 10\%$

What's on your circuit? : Resistors



4 Band Resistor Color Coding



COLOR	1ST BAND	2ND BAND	MULTIPLIER	TOLERANCE
BLACK	0	0	x1 Ω	
BROWN	1	1	x10 Ω	$\pm 1\%$
RED	2	2	x100 Ω	$\pm 2\%$
ORANGE	3	3	x1000 Ω	
YELLOW	4	4	x10000 Ω	
GREEN	5	5	x100000 Ω	$\pm 0.5\%$
BLUE	6	6	x1000000 Ω	$\pm 0.25\%$
VIOLET	7	7	x10000000 Ω	$\pm 0.10\%$
GREY	8	8		$\pm 0.05\%$
WHITE	9	9		
GOLD			0.1	$\pm 5\%$
SILVER			0.01	$\pm 10\%$

Difficult to see on projector:
this says "x1 Ω ", not "x10"

Poll Time! What color is a 100 ohm resistor?

4 Band Resistor Color Coding



COLOR	1ST BAND	2ND BAND	MULTIPLIER	TOLERANCE
BLACK	0	0	$\times 1\Omega$	
BROWN	1	1	$\times 10\Omega$	$\pm 1\%$
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1. black-brown-red
2. brown-black-brown
3. brown-black-red
4. brown-black-black

Poll Time! What color is a 100 ohm resistor?

4 Band Resistor Color Coding



COLOR	1ST BAND	2ND BAND	MULTIPLIER	TOLERANCE
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GOLD			0.1	$\pm 5\%$
SILVER			0.01	$\pm 10\%$

1. black-brown-red
2. **brown-black-brown**
3. brown-black-red
4. brown-black-black

Poll Time! What color is a 100K resistor? (100 kilo-ohms, so 100,000 ohms)



4 Band Resistor Color Coding				
COLOR	1ST BAND	2ND BAND	MULTIPLIER	TOLERANCE
BLACK	0	0	$\times 1\Omega$	
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1. brown-black-red
2. brown-black-brown
3. brown-black-yellow
4. brown-black-white

Poll Time! What color is a 100K resistor? (100 kilo-ohms, so 100,000 ohms)



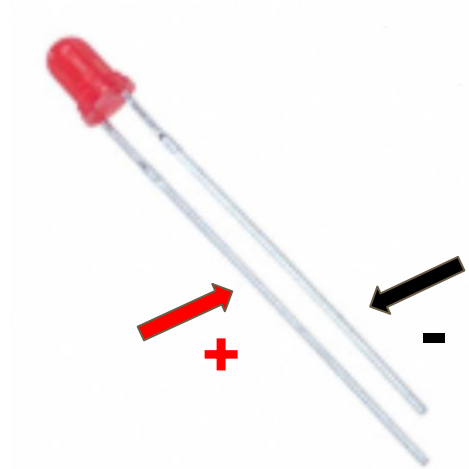
4 Band Resistor Color Coding				
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SILVER			0.01	$\pm 10\%$

1. brown-black-red
2. brown-black-brown
- 3. brown-black-yellow**
4. brown-black-white

Light Emitting Diode (LED)

When a sufficient potential difference (voltage) is placed across its terminals, the LED emits light!

Direction matters!



Ambient Light Sensor

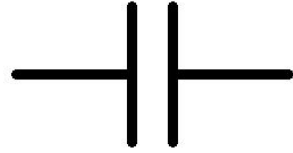


It behaves like a resistor and the current passing through it depends on how much light there is around it!

Direction matters! **Note: Polarity is opposite LED's**

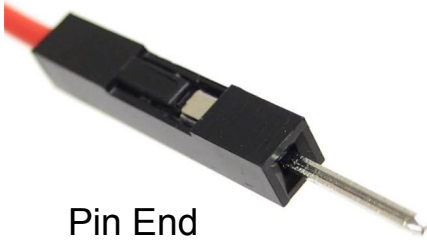


Capacitors

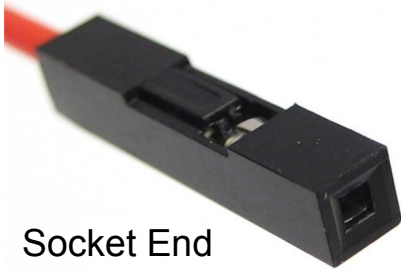


They store your charge!
Called capacitors because
they have a set capacity (in
Farads)

Wires/Jumpers



Pin End



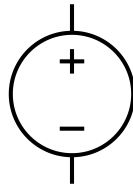
Socket End



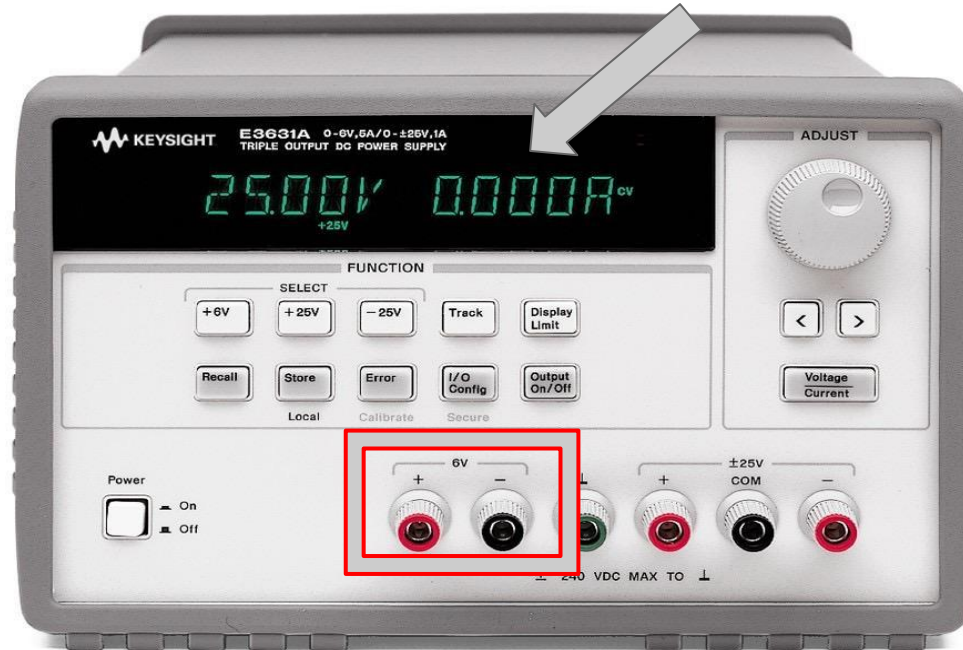
IMPORTANT: we use pin/socket terminology for wiring. You may encounter male/female in documentation or in industry.



Voltage Source

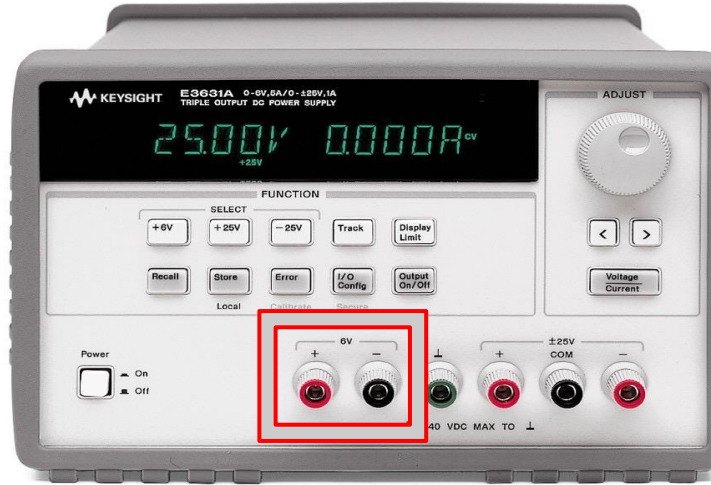
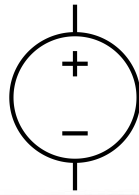


IMPORTANT: Always keep current limited @ 0.1 A limit

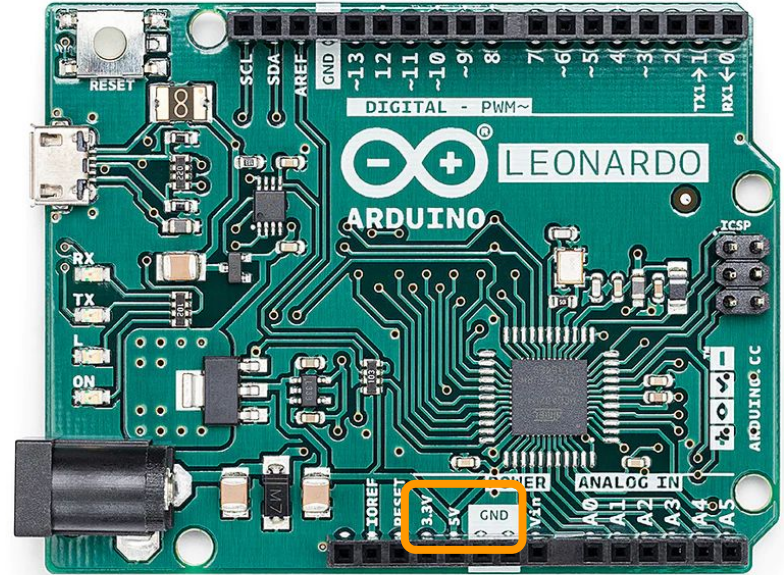


PSU cables
are hanging
on back wall

Voltage Source



Power Supply Unit (PSU)



We will be using the Arduino as well as the PSU as our voltage source. The 3V3 and GND pins on the Arduino are the + and - terminals of the voltage source respectively

Simple Circuit: The Theory

- Components
- Nodes
 - Points in circuit where circuit elements meet
 - Wire between components are considered part of one node
- We know you don't know much about circuits yet; we've given you very detailed instructions on how to build the circuit in the Jupyter Notebook. Please read the Notebook carefully.

Simple Circuit: The Theory™

- Components (Resistors, LEDs, Capacitors)
- Nodes
 - Point in circuit where circuit elements meet
 - Wire between components are considered part of one node



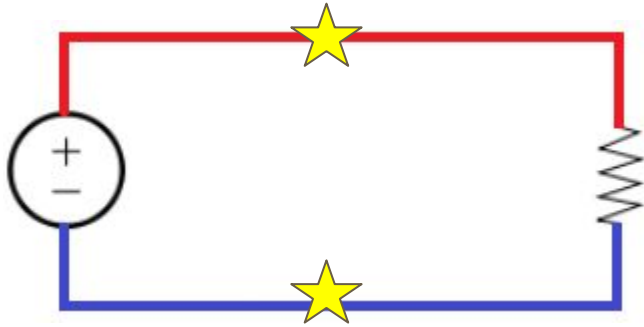
What components?

How many nodes?

Where are these nodes?

Simple Circuit: The Theory™

- Components (Resistors, LEDs, Capacitors)
- Nodes
 - Point in circuit where circuit elements meet
 - Wire between components are considered part of one node



What components?

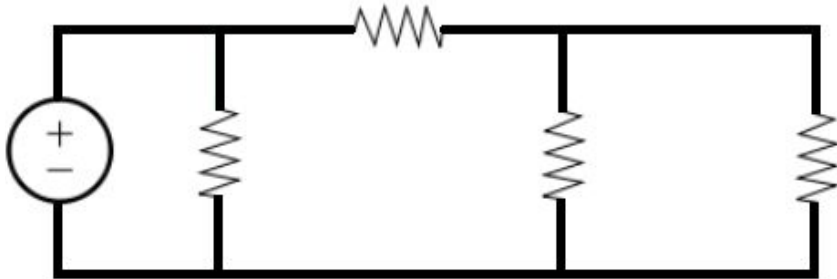
Voltage source, resistor

How many nodes? **2**

Where are these nodes?

Simple Circuit: The Theory™

- Components (Resistors, LEDs, Capacitors)
- Nodes
 - Point in circuit where circuit elements meet
 - Wire between components are considered part of one node



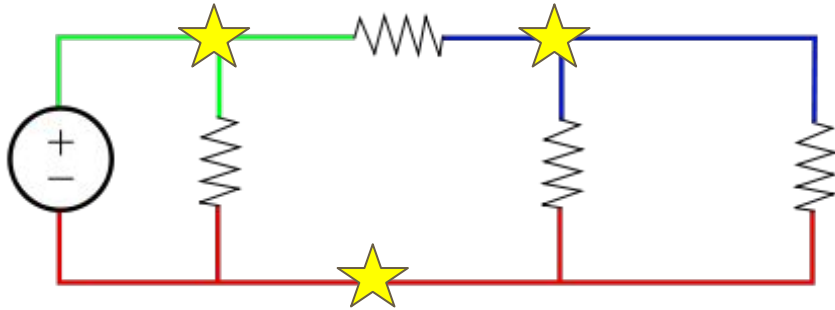
What components?

How many nodes?

Where are these nodes?

Simple Circuit: The Theory™

- Components (Resistors, LEDs, Capacitors)
- Nodes
 - Point in circuit where circuit elements meet
 - Wire between components are considered part of one node

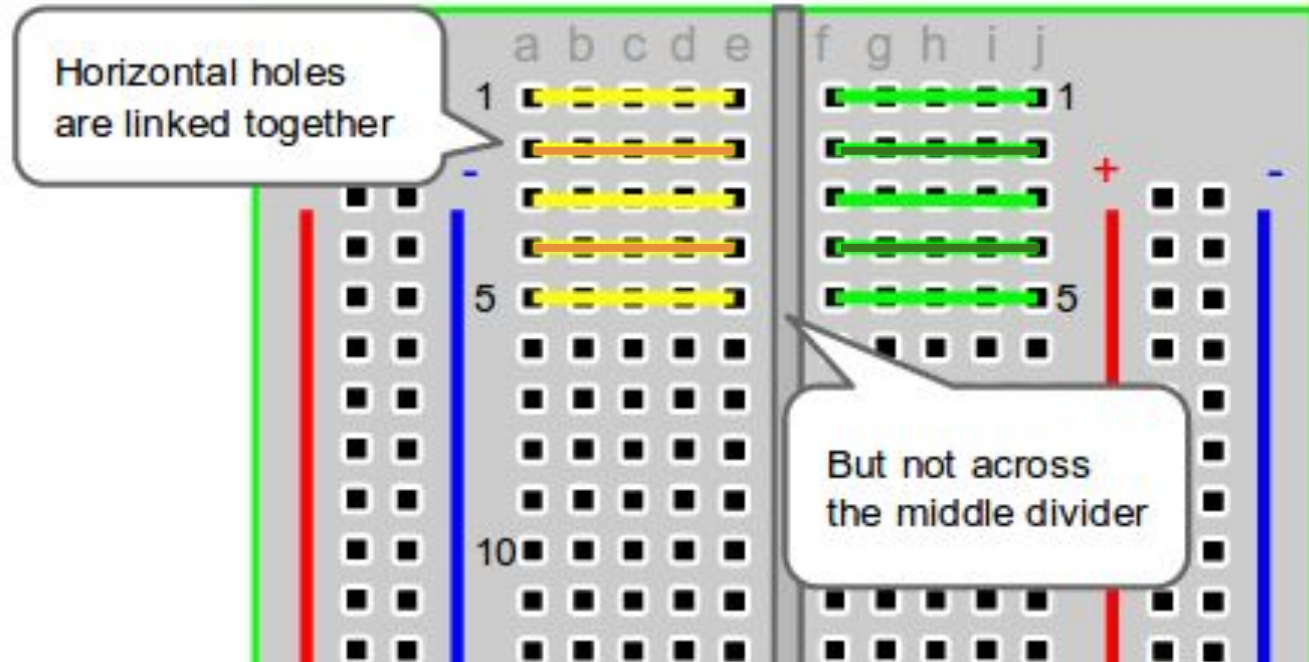


What components? **Same**

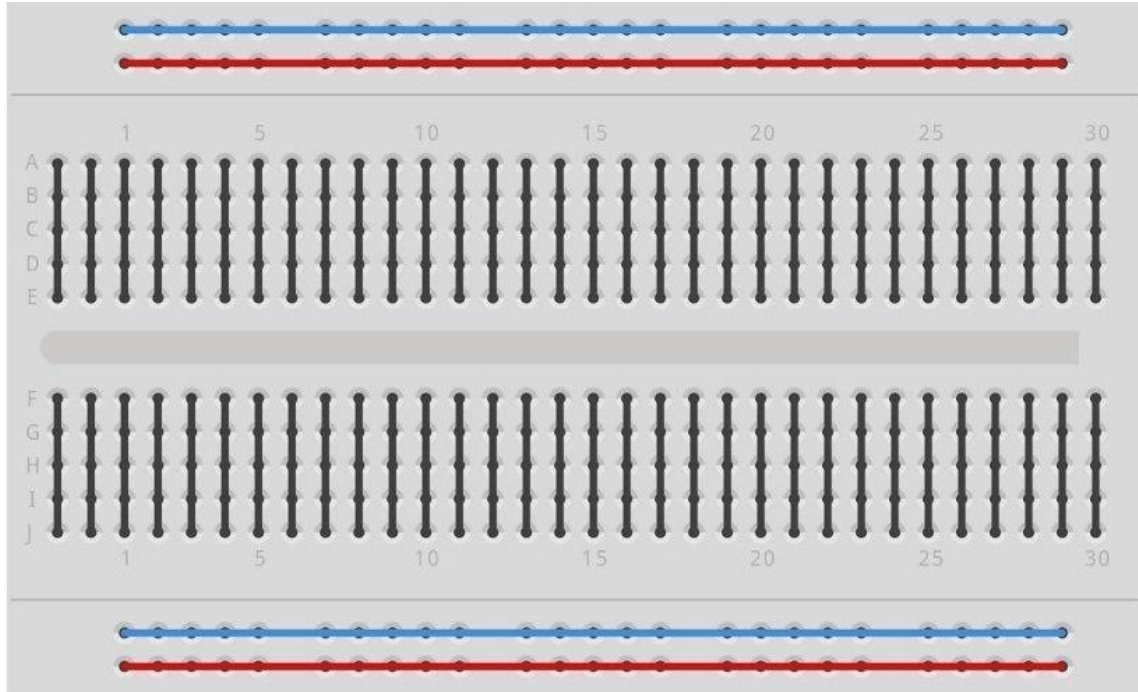
How many nodes? **3**

Where are these nodes?

Breadboard

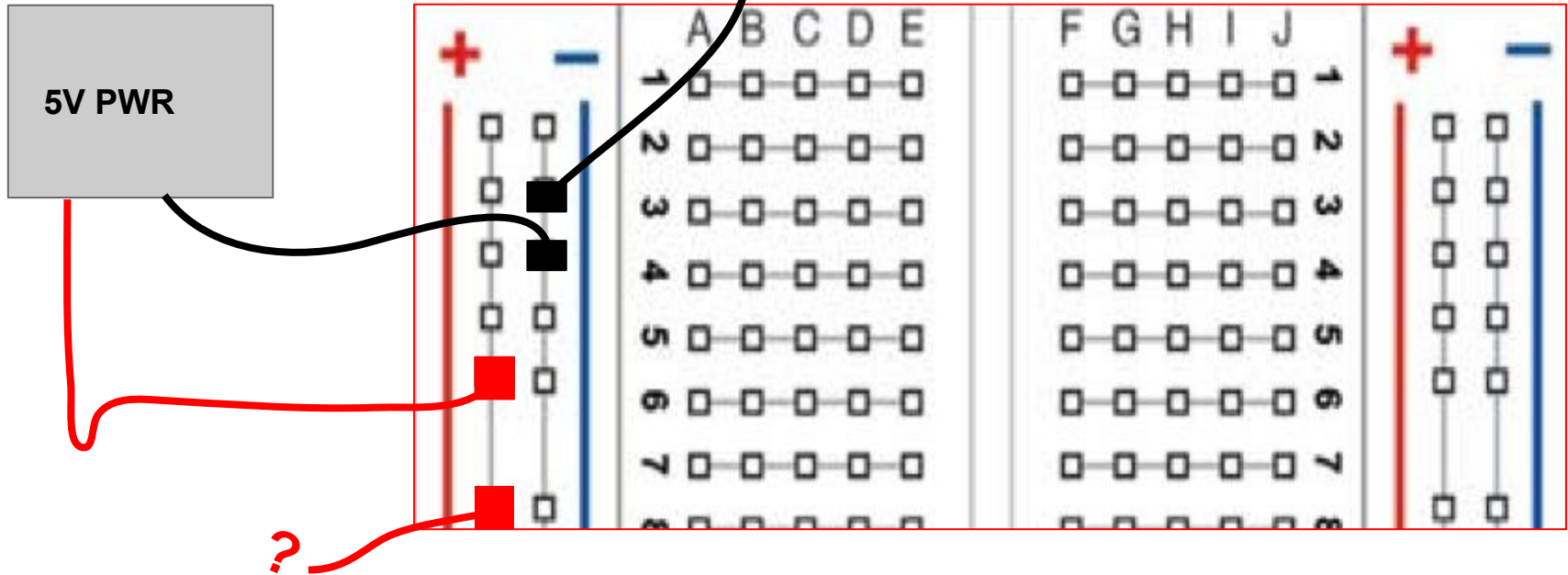
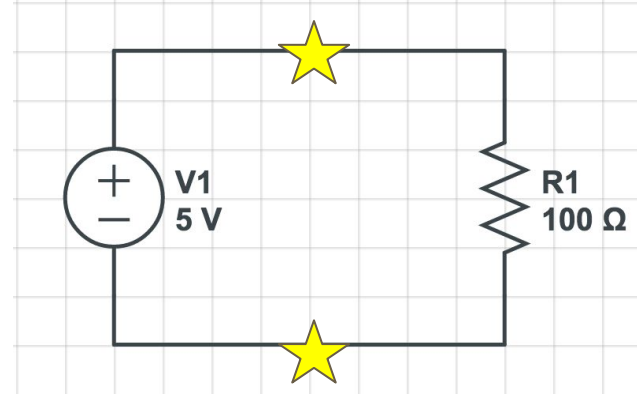


Breadboard



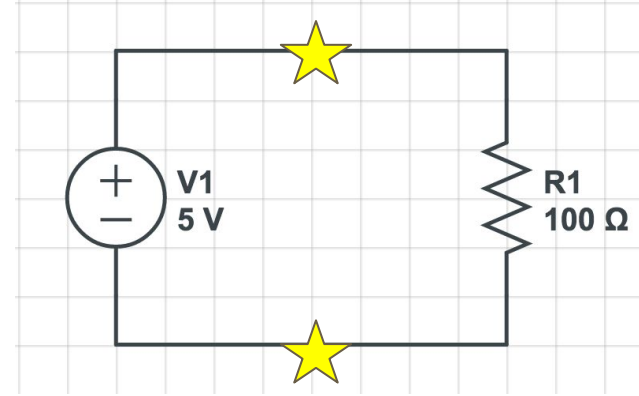
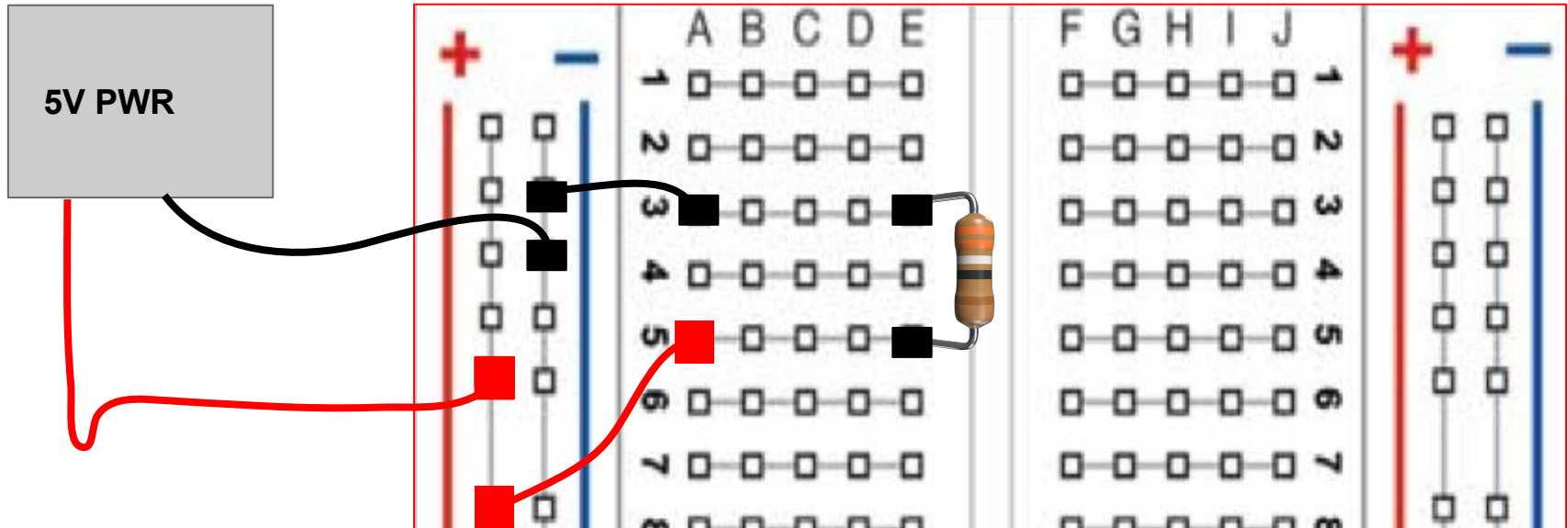
Breadboard Do's and Don't's

How do we make this circuit? →



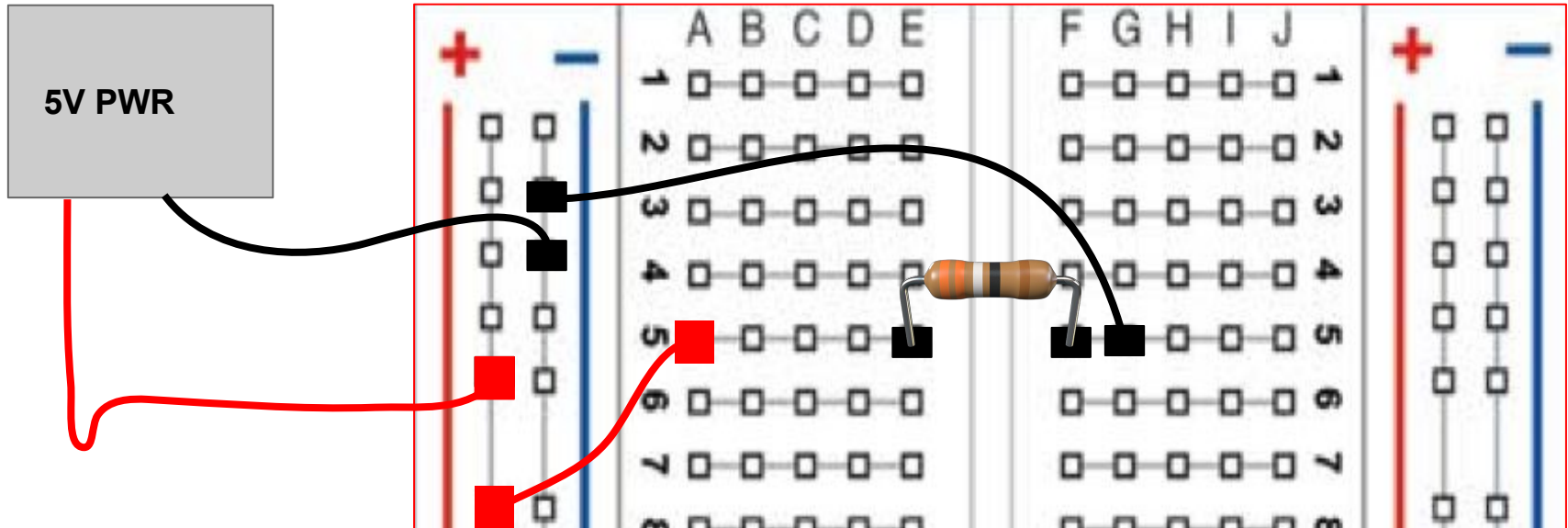
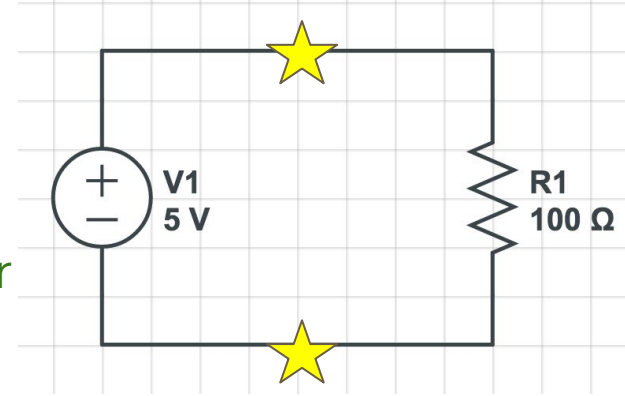
Breadboard Do's and Don't's

- ✓ **Do** plug component's ends into two different rows - separate nodes



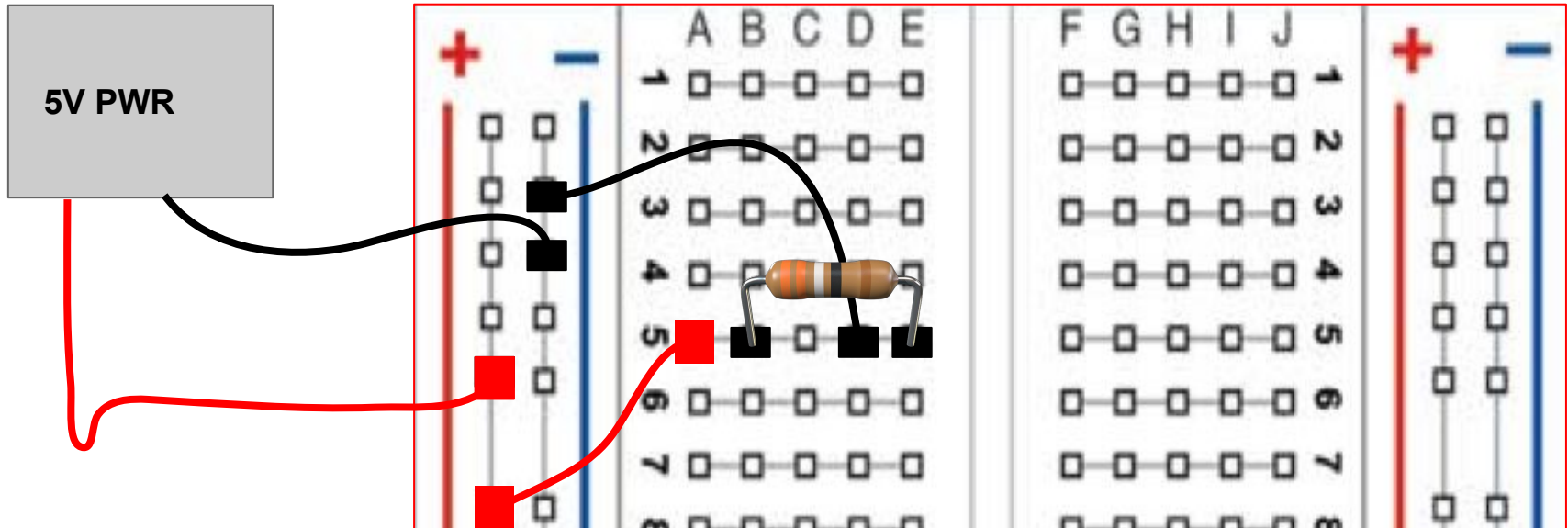
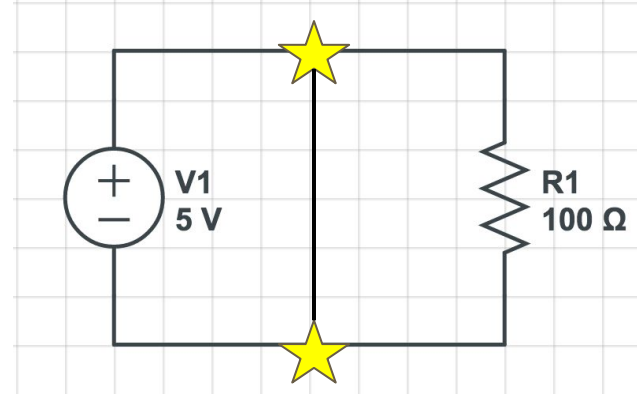
Breadboard Do's and Don't's

- ✓ **Do** plug components across the gap in your breadboard - A-E and F-J are separate



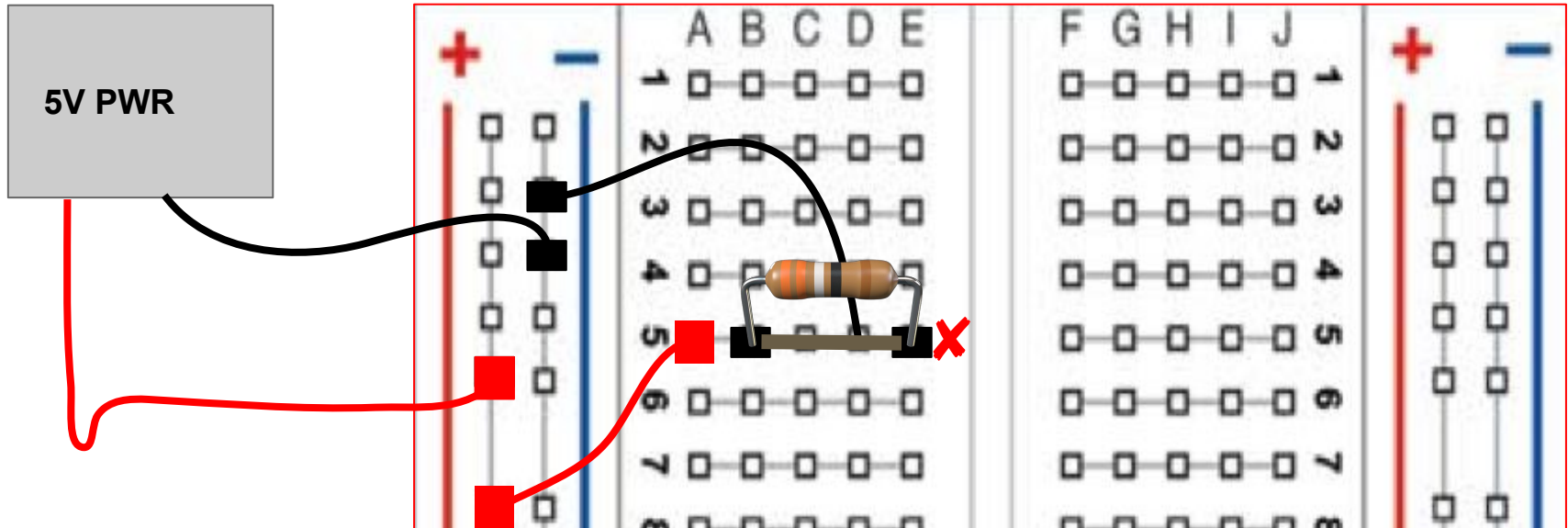
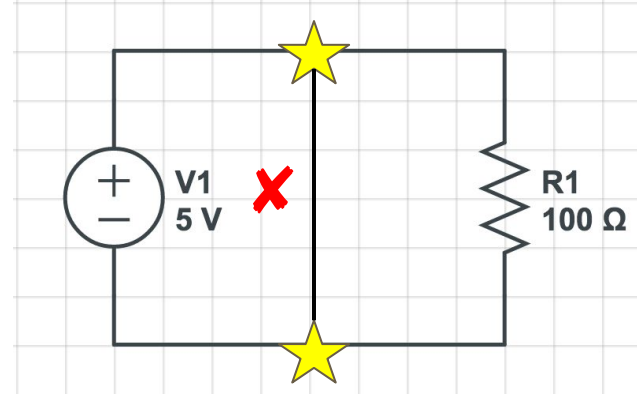
Breadboard Do's and Don't's

Is this okay? If there is an error, where?

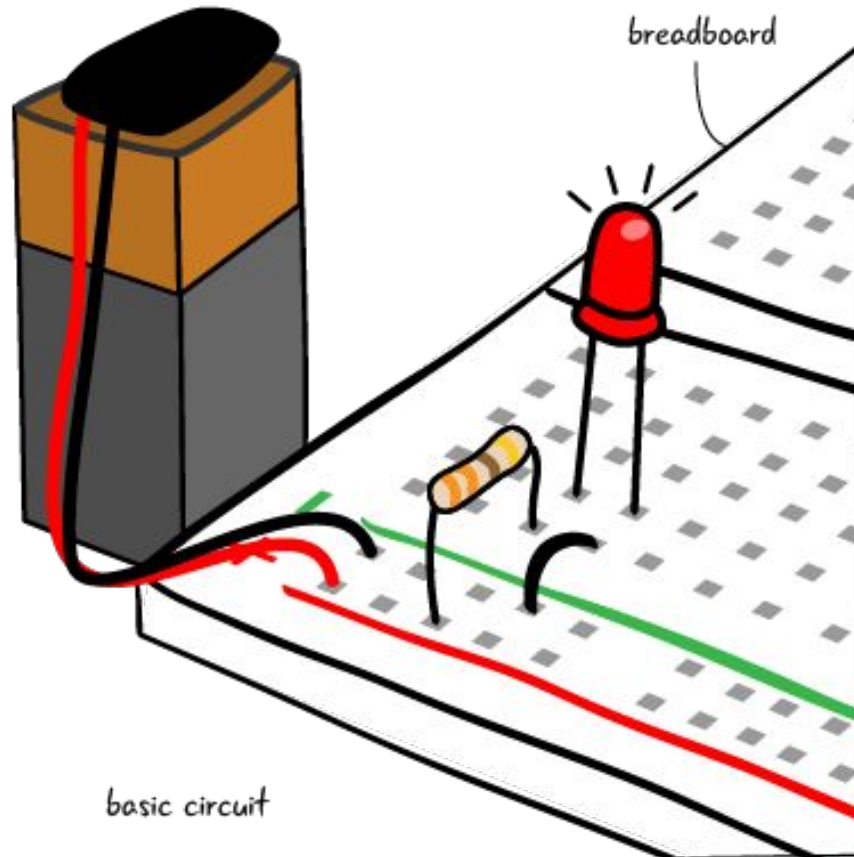


Breadboard Do's and Don't's

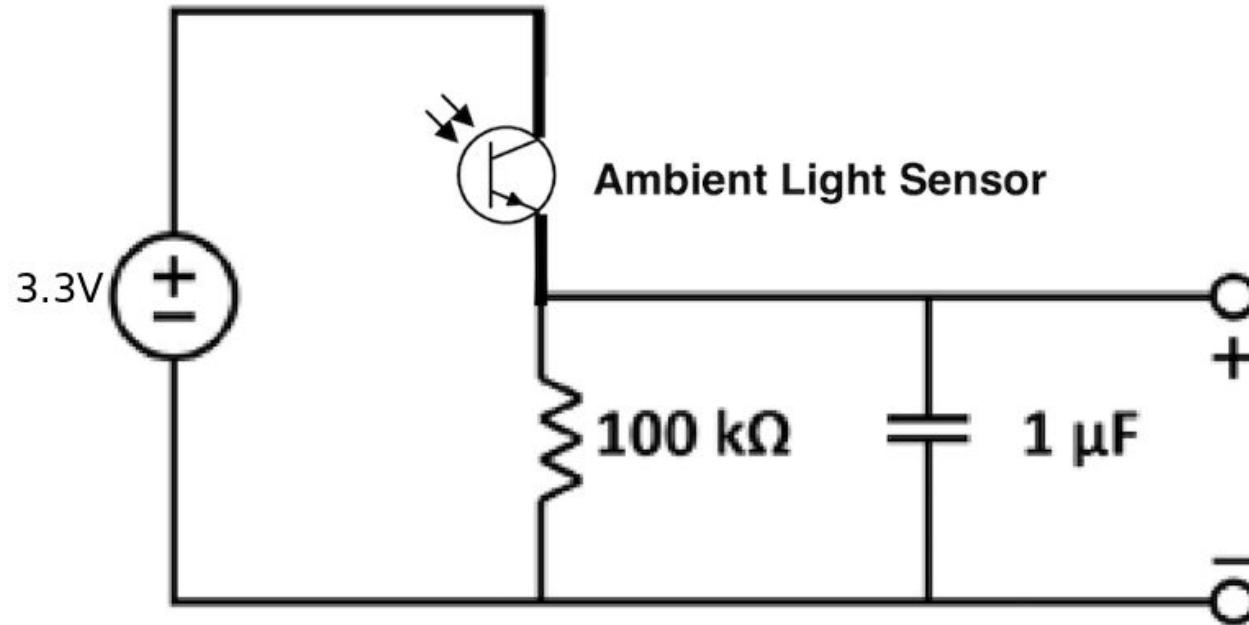
✗ **Do not** plug both ends of component into the same row! This creates a short



Breadboarding Color Convention

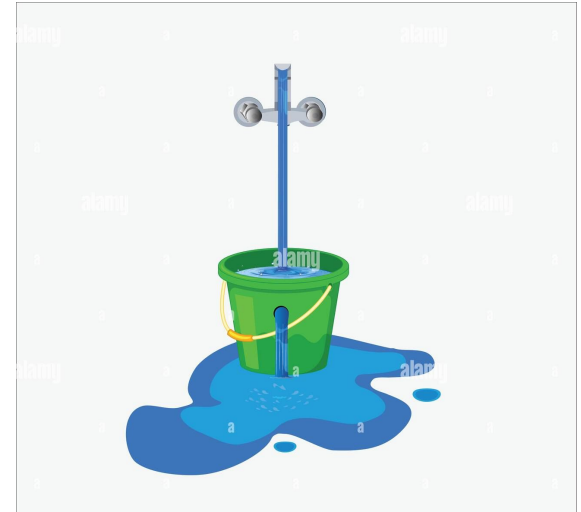
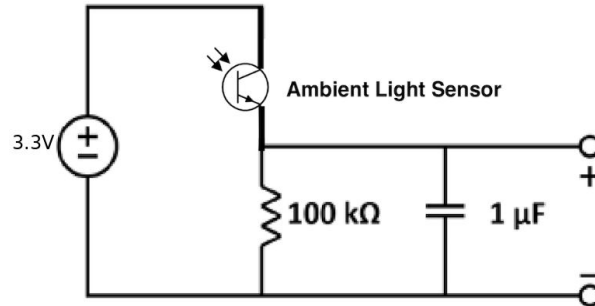
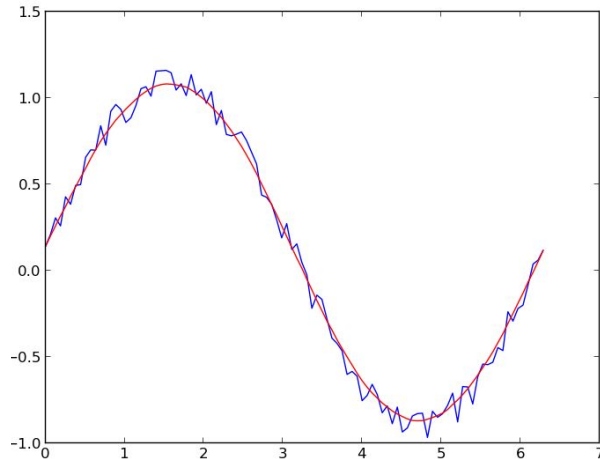


Light-detecting Circuit



Why the Capacitor?

- The capacitor acts like a bucket of charge – if the input instantaneously increases or decreases, it'll adjust the output flow to compensate.
- This results in reducing noise and curve smoothing!



FAQ: Running Jupyter

- Download the ZIP file from the course website
- Extract the files
- Click on the launch file (Launch.bat) to bring up Jupyter on a local server
- You should see the lab files on a browser window that opens up automatically
- Click on the .ipynb file to start working on the lab!
- If images in the notebook don't show up, save your work and reopen the notebook/reload the page (F5)

FAQ: Safety

- Make sure current limit of power supply is set to 0.1A
- Turn PSU **output off** while building your circuit
- Keep voltage source leads from Arduino Leonardo to breadboard disconnected while building your circuit
 - Socket ends can stay connected to the Arduino

FAQ: Lab Equipment

- Oscilloscope and PSU probes are on the back wall
- Make sure you are using the **correct resistors**
(Brown Black Yellow Gold for light sensor)



- Make sure your ambient light sensor is in the right direction
- Before leaving, please return the wires, clean up your stations, and sign out of the computers

FAQ: Working In Pairs

- Complete the lab in PAIRS, do ONE setup and notebook per groups.
- Speak to the staff if you do not have a lab partner.

FAQ: Check-off and leaving the lab

- DON'T LEAVE THE LAB OR PACK UP YOUR CIRCUIT WITHOUT BEING CHECKED OFF FIRST.
- To get credit for the lab:
 - Fill out the checkoff form.
 - Both partners must fill out individual forms. You can use the same signed-in email but make sure to submit the **correct SID**.
 - Wait for a staff member (TA/UCS1) to complete the check-off.
 - **Return Arduino Leonardo** to a staff member
 - Clean up your lab station.
 - Do not turn off/shut down the computers when you are done. You may switch off the monitors.

Reminders

- Unzip the lab folder.
- Safety: 0.1A PSU limit, no food or drink.
- Today's resistor:



- Be careful with lab equipment. If you are unsure about something, ask.
- Check-off, **return Arduino** and clean up the lab station to get credit.
- (Optional) Student Feedback:
<https://tinyurl.com/fb-student-fa23>

Checkoff Form:



Feedback Form:

